



Kotlin

Training Assignment

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RECORD OF CHANGES

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Assignment 1: Basic OOP in Kotlin

Create a Vehicle class which is an abstract class that represents the means of transport. Define attributes such as license plate, year of manufacture, and price. Provides a constructor to initialize properties and **methods to access information and calculate the total price**.

Create two subclasses Car and Motorcycle that inherit from Vehicle class. Define specific properties for each vehicle type, along with the corresponding constructor and methods.

Create a VehicleManager class to manage the list of vehicles. This class will have a list of Vehicle objects and provide methods to **add vehicles, display the list, and calculate the total price**.

In the main function or a similar method, create a VehicleManager object and do some operations that add vehicles, display a list, and calculate the total price.

Technical Requirements:

- Kotlin OOP(using abstract class or interface)

Assignment 2: Advanced OOP in Kotlin

Build a contact management application. Each contact can contain many types of contact information such as phone numbers, email addresses, social media accounts

Technical Requirements:

Generics and Sealed Classes

- Create a Sealed Class ContactInformation to represent the type of contact information.
- Create subclasses of ContactInformation like PhoneNumber, EmailAddress, SocialMediaAccount, etc.
- Use Generics to create a Contact<T: ContactInformation> class that represents a phonebook with specific contact information. This class will have a property named list T.

Object Expressions and Object Declarations:

- Use Object Expressions to create a sample directory with some contact information.
- Create an Object Declaration as ContactManager to manage contacts.
- In ContactManager, use the sample contacts created with Object Expressions.

Program implementation:

- In the main function, use ContactManager to add and display contacts.
- Displays a list of contacts and contact information for each contact.

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